Course notes

Vector calculation

Chasles' relationship $\overrightarrow{AB} + \overrightarrow{BC} = \overrightarrow{AC}$, the sum of vectors and the product by a real extend
Two non-zero vectors are colinear if one is a multiple of the other.
Three vectors are coplanar if they belong to the same plane
Yes, I am. $\vec{u}, \vec{v}$ are not colinear, $\vec{u}, \vec{v}, \vec{w}$ are coplanar if and only if $\vec{w} = \alpha\vec{u} + \beta\vec{v}$ for some real α, β.

Right and plan

The right through A of a non-zero master vector $\vec{u}$ is all points M such that $\overrightarrow{AM} = t\vec{u}$ for a real t. The plan passing through A and directed by two non-colonial vectors $\vec{u}, \vec{v}$ is all points M such that $\overrightarrow{AM} = s\vec{u} + t\vec{v}$ for real s, t.

Three non-aligned items define a plan. $A, B, C, D,$ A, B, C Non-Aligned coplanar if and only if $\overrightarrow{AD}$ is linear combination of $\overrightarrow{AB}$ and $\overrightarrow{AC}$.

Positions

Two straight lines are confused, strictly parallel, seizing or not coplanar.
colinear directors characterize parallel straights in the broad sense.
do not provide the intersection. A straight and a plane: straight included in the plane, parallel and disjointed,
Two planes: combined, strictly parallel, or slatting according to one point.
Right.

Example in a cube. In ABCDEFGH , the faces $ABCD$ and $EFGH$ are parallel and edges AE, BF, CG, DH The lines match. (AB) and (CG) are neither parallel nor secant: they are non-coplanar.

Base and coordinates

A base of space is a family of three non-coplanar vectors.
unique way $x\vec{\imath} + y\vec{\jmath} + z\vec{k}$. A reference is made up of an origin O and a base.

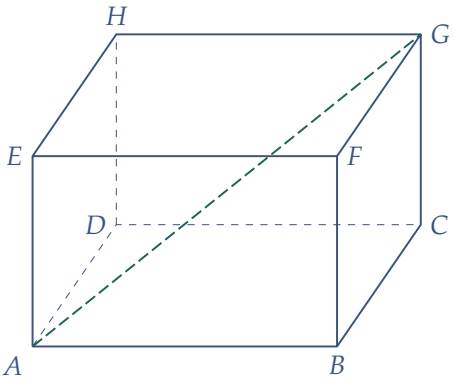
$$\overrightarrow{AB} = (x_B - x_A, y_B - y_A, z_B - z_A), \qquad I = \left(\frac{x_A + x_B}{2}, \frac{y_A + y_B}{2}, \frac{z_A + z_B}{2}\right).$$

In an orthonormal coordinate system, $\|\vec{u}\| = \sqrt{x^2 + y^2 + z^2}$ and $AB = \sqrt{(x_B - x_A)^2 + (y_B - y_A)^2 + (z_B - z_A)^2}$.

Example. In an orthonormal coordinate system, $A(1, 0, 2)$, $B(3, -1, 4)$, $\overrightarrow{AB} = (2, -1, 2)$, $AB = 3$ and midpoint is $(2, -\frac{1}{2}, 3)$.

Test a linear combination

Write the three co-ordinates. Determine coefficients from two equations
When possible, then check the third. A contradiction proves non-membership;
We don't remove the awkward equation.



$[AG]$ is a diagonal of space, plotted in green dots.
The hidden edges are dotted blue; D and F are not sure (AG).

Build a reliable representation

10.1 Combinations and any base

First use: S3 and S6.

Definition · Decompose a vector

A base of space is a triplet of non-coplanar vectors.
a unique way as a combination of these three vectors. A base is not necessarily
orthogonal or formed by unit vectors.

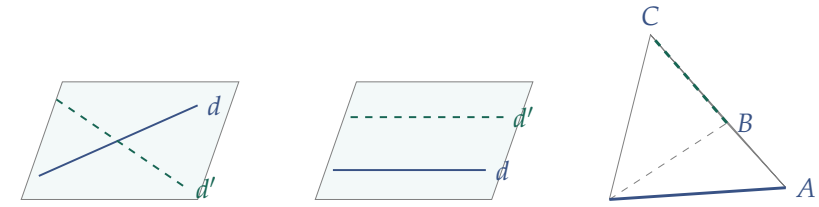
Example solved · In a tetrahedron

In a tetrahedron not flattened $OABC$, posisons $\vec{u} = \overrightarrow{OA}$, $\vec{v} = \overrightarrow{OB}$, $\vec{w} = \overrightarrow{OC}$. If I is the midpoint of $[AB]$, then $\overrightarrow{OI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v}$. So $\overrightarrow{CI} = \frac{1}{2}\vec{u} + \frac{1}{2}\vec{v} - \vec{w}$. These equalitys use Chasles and the midpoint, not lengths or angles.

A1 · Immediate application With the same base, J is the midpoint of $[OC]$ and K checks $\overrightarrow{OK} = 2\vec{u} - \vec{v}$. Express $\overrightarrow{JK}$ then $\overrightarrow{IK}$.

10.2 See not enough to conclude

First use: S12.



pruning, same plane parallel, same plane non-coplanar edges

Property · Two-right positions

Two lines are either coplanar (secting or parallel) or non-coplanar.
non-collinear managers exclude parallelism, but do not prove an intersection.

Example Solved · Two non-coplanar lines

In tetrahedron $OABC$, the right (OA) and (BC) are not coplanar. If a plan
both contained, it would contain O, A, B, C, which would contradict the tetrahedron hypothesis
not flattened. An intersection of their lines on the sheet would only be a projection effect.

A2 · Read and argue On the three diagrams, name the position of the straights.
geometrically accompanying the drawing to allow this answer ? Give an example of straight lines not
coplanar in a cube ABCDEFGH .

A3 · Refuter "Two straight lines whose guiding vectors are not colinear are pruning."
Refuse with tetrahedron, then add a hypothesis that makes the statement true.

10.3. Characterize a plane by two directions

First use: S12 and S15.

Example resolved · Membership and uniqueness

On a base, be $\vec{u} = (1;0;1)$, $\vec{v} = (0;1;1)$ and $\overrightarrow{AM} = (2;3;5)$. We're looking. λ, μ such that $\overrightarrow{AM} = \lambda\vec{u} + \mu\vec{v}$. The first two coordinates require $\lambda = 2$, $\mu = 3$ and the third check $5 = 2 + 3$. Item M belongs to the point plan A and Directions $\vec{u}, \vec{v}$. For $(2;3;4)$, the The last tie fails: the point does not belong to it.

A4 · Partial guidance The vectors $\vec{u} = (1;1;0)$ and $\vec{v} = (0;1;1)$ Are they colinear? value of m the vector $(2;m;3)$ Is it a combination of $\vec{u}$ and $\vec{v}$?

A5 · Correct an error On any basis, a student calculates the norm of the vector of Coordinated $(1;1;0)$ by $\overline{2}$. Explain the missing hypothesis. Give an example with two unit vectors at an angle of 60 $^\circ$.

10.4. Moving from solid to calculation

First use: S17.

Example Solved · Middles and Parallelism

If I and J are the midpoints of [OA] and [OB], $\overrightarrow{IJ} = \overrightarrow{IO} + \overrightarrow{OJ} = -\frac{1}{2}\overrightarrow{OA} + \frac{1}{2}\overrightarrow{OB} = \frac{1}{2}\overrightarrow{AB}$. The rights (IJ) and (AB) therefore have colinear directions. This calculation is valid in any triangle not flattened.

A6 · Choice of strategy In tetrahedron $OABC$, I is the midpoint of $[OA]$, J that of $[OB]$ and K the of $[OC]$. Demonstrate that plans (IJK) and (ABC) are parallel, then that O does not belong to (IJK). Do not use Cartesian equation.

A first step: Compare $\overline{IJ}$ to $\overrightarrow{AB}$ and $\overrightarrow{IK}$ to $\overrightarrow{AC}$. For membership, decompose in $(\overrightarrow{OA}, \overrightarrow{OB}, \overrightarrow{OC})$.

Wait a minute at the courthouse. Describe the tetrahedron and the three midpoints without showing the figure. must be able to place them. Specify the segments rather than the "left-hand point".

Review postponed · S51 and then S70 The same plane will be described by a normal vector, then an equation. Vector definition remains the means of controlling the calculation.

Research activity

Three midpoints in a cube · 40 min.

The cube is considered ABCDEFGH shown above. $\vec{\imath} = \overrightarrow{AB}$, $\vec{\jmath} = \overrightarrow{AD}$ and $\vec{k} = \overrightarrow{AE}$. The points I, J and K are the respective circles of $[AB]$, $[AD]$ and $[AE]$. We note O the center of the cube.

1. Express $\overline{IJ}$ and $\overrightarrow{IK}$ in base $(\vec{\imath}, \vec{\jmath}, \vec{k})$.

2. Demonstrate points I, J and K are not aligned.

3. Determine whether O belongs to the plan (IJK). The answer will be justified by whether $\overrightarrow{IO}$ is a linear combination of $\overline{IJ}$ and $\overrightarrow{IK}$.

4. Why does the single reading of the figure not allow us to demonstrate belonging or non-membership O on the plan (IJK) ?